

VLAN Trunking and Access Ports

Task 1: In Cisco Packet Tracer, configure two switches (Switch1 and Switch2) and connect them with a crossover cable. Set the FastEthernet0/1 port on both switches as trunk ports using the 802.1Q protocol.

Switch0 and Switch1 (Cisco 3560-24 multilayer switches) were connected via their Fa0/1 ports. The following commands were used to configure the trunk on both switches:

```
Switch(config)# interface fastEthernet 0/1
```

```
Switch(config-if)# switchport trunk encapsulation dot1q
```

```
Switch(config-if)# switchport mode trunk
```

The trunk was verified using the show interface trunk command on both switches:

Output Modifiers			
<cr>		Switch(config-if)#no shut	
Switch#show interface trunk		Switch(config-if)#do wr	
Port	Mode	Building configuration...	
Fa0/1	on	(OK)	
	Encapsulation	Switch(config-if)#end	
	802.1q	Switch#	
	Status	%SYS-5-CONFIG_I: Configured from console by console	
	trunking	Switch#show interface trunk	
	Native vlan	Port	Mode
	1	Fa0/1	on
Port	Vlans allowed on trunk	Encapsulation	Status
Fa0/1	1-1005	802.1q	trunking
		Native vlan	1
Port	Vlans allowed and active in management domain		
Fa0/1	1	Port	Vlans allowed on trunk
		Fa0/1	1-1005
Port	Vlans in spanning tree forwarding state and not pruned		
Fa0/1	1	Port	Vlans allowed and active in management domain
Switch#		Fa0/1	1
		Port	Vlans in spanning tree forwarding state and not pruned
		Fa0/1	1
		Switch#	

Fig 1.1 - show interface trunk output on Switch0 (left) and Switch1 (right), confirming Fa0/1 is trunking with 802.1q encapsulation and native VLAN 1.

Task 2: Create three VLANs (10, 20, 30) on both switches and assign FastEthernet0/2, 0/3, and 0/4 on Switch1 to VLAN 10, 20, and 30 respectively as access ports.

VLANs 10, 20 and 30 were created on both switches, and access ports were assigned on Switch0 using the following commands:

```
Switch(config)# vlan 10
```

```
Switch(config)# vlan 20
```

```
Switch(config)# vlan 30
```

```
Switch(config)# interface fastEthernet 0/2
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 10
```

```
Switch(config)# interface fastEthernet 0/3
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 20
```

```
Switch(config)# interface fastEthernet 0/4
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 30
```

The configuration was verified with show vlan brief on both switches:

```

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11,
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15,
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19,
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23,
                                           Fa0/24
10   VLAN0010                active    Gig0/1, Gig0/2
                                           Fa0/2
20   VLAN0020                active    Fa0/3
30   VLAN0030                active    Fa0/4
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#

```

Fig 2.1 - show vlan brief confirming VLAN 10 (Fa0/2), VLAN 20 (Fa0/3), and VLAN 30 (Fa0/4).

Multilayer Switch1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```

Switch(config-if)#do wr
Building configuration...
[OK]
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan ?
  brief    VTP all VLAN status in brief
  id       VTP VLAN status by VLAN id
  name     VTP VLAN status by VLAN name
  <cr>

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11,
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15,
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19,
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23,
                                           Fa0/24
10   VLAN0010                active    Gig0/1, Gig0/2
                                           Fa0/2
20   VLAN0020                active    Fa0/3
30   VLAN0030                active    Fa0/4
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#

```

Fig 2.2 - Identical VLAN configuration verified on Switch1.

Task 3: Connect three PCs to Switch1 on ports assigned to VLAN 10, 20, and 30. Verify that PCs in different VLANs cannot communicate with each other, but can communicate with other PCs in the same VLAN.

The topology was extended to six PCs across both switches (PC0-PC2 on Switch0, PC3-PC5 on Switch1), one per VLAN per switch, to test both same-switch and cross-switch (trunk) VLAN behaviour:

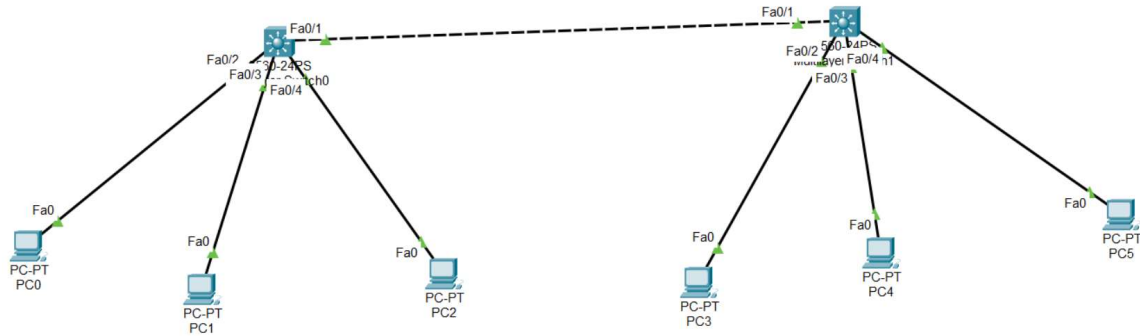


Fig 3.1 - Final topology showing PC0-PC5 distributed across VLANs 10, 20 and 30 on both switches.

PC-to-VLAN-to-IP mapping used for testing:

PC	Switch	VLAN	IP Address
PC0	Switch0	10	192.168.1.11
PC1	Switch0	20	192.168.1.12
PC2	Switch0	30	192.168.1.13
PC3	Switch1	10	192.168.1.14
PC4	Switch1	20	192.168.1.15
PC5	Switch1	30	192.168.1.16

Ping results:

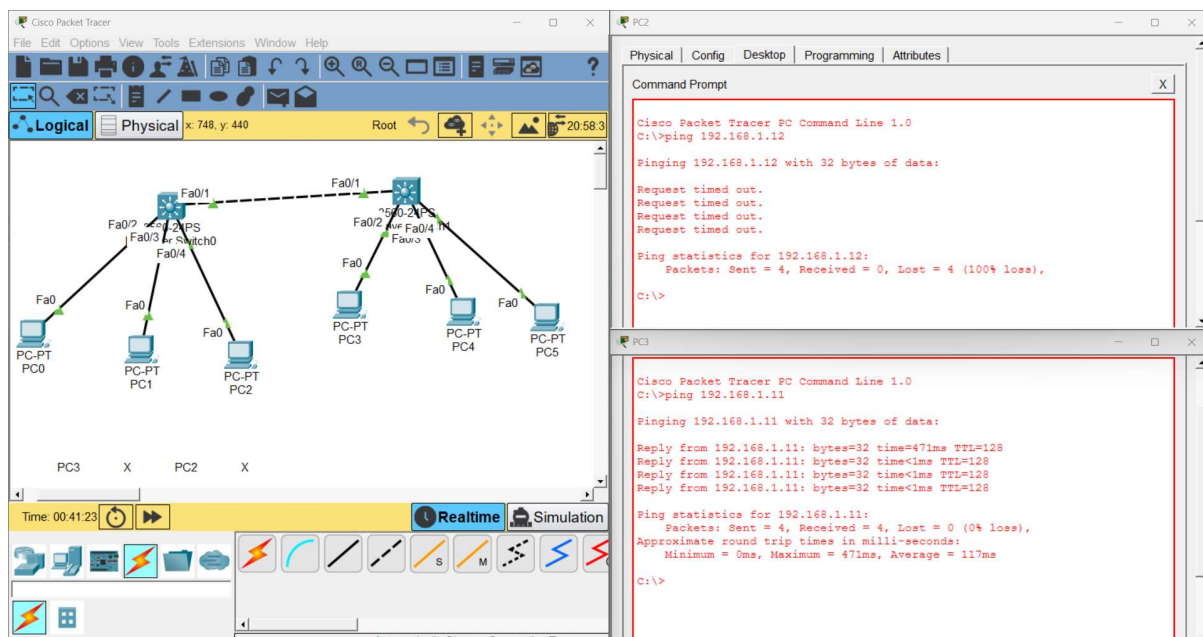


Fig 3.2 - PC2 (VLAN 30) to 192.168.1.12 / PC1 (VLAN 20) failed (100% loss - different VLANs). PC3 (VLAN 10) to 192.168.1.11 / PC0 (VLAN 10) succeeded (0% loss) despite being on different switches, proving the trunk correctly carries VLAN 10 traffic.

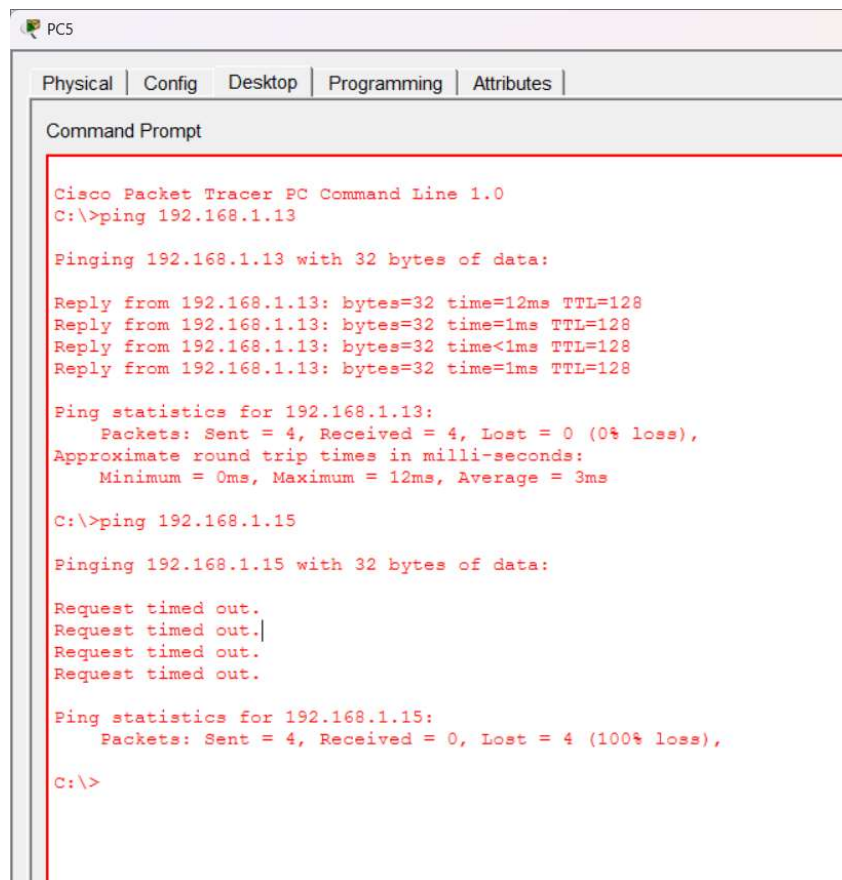
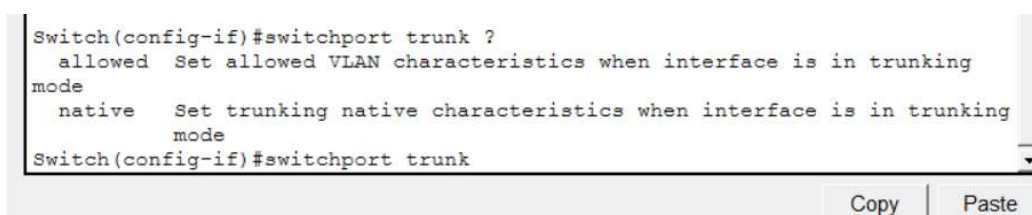


Fig 3.3 - PC5 (VLAN 30) to 192.168.1.13 / PC2 (VLAN 30) succeeded across the trunk. PC5 to 192.168.1.15 / PC4 (VLAN 20) failed - different VLANs blocked.

Conclusion: PCs in the same VLAN can communicate freely, even across the trunk link between the two switches, while PCs in different VLANs cannot communicate at all - regardless of which physical switch they are connected to. This confirms that VLAN membership, not physical switch location, determines Layer 2 reachability, and that the 802.1Q trunk is correctly tagging and forwarding VLAN traffic between switches.

Task 4: Change the trunk encapsulation on the trunk link between Switch1 and Switch2 from 802.1Q to ISL, and test if VLAN traffic still passes between the switches.

This task was attempted using the command 'switchport trunk encapsulation isl' across four different switch models (Cisco 2960, 2950-24, 2950T-24, and 3560-24) to confirm the finding was not model-specific.



```
Switch(config-if)#switchport trunk ?
  allowed  Set allowed VLAN characteristics when interface is in trunking
mode
  native   Set trunking native characteristics when interface is in trunking
mode
Switch(config-if)#switchport trunk
```

Fig 4.1 - Cisco 2960: 'switchport trunk' offers only 'allowed' and 'native' sub-options; no 'encapsulation' keyword exists at all.

```
IOS Command Line Interface
Motherboard revision number: A0
Model number: WS-C2950-24
System serial number: FHK0610Z0WC

Cisco Internetwork Operating System Software
IOS (tm) C2950 Software (C2950-I6Q4L2-M), Version 12.1(22)EA4, RELEASE
SOFTWARE(fc1)
Copyright (c) 1986-2005 by cisco Systems, Inc.
Compiled Wed 18-May-05 22:31 by jharirba

Press RETURN to get started!

Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/0
%Invalid interface type and number
Switch(config)#int fa0/1
Switch(config-if)#switchport ?
    access      Set access mode characteristics of the interface
    mode        Set trunking mode of the interface
    nonegotiate  Device will not engage in negotiation protocol on this
                  interface
    port-security Security related command
    priority     Set appliance 802.1p priority
    protected    Configure an interface to be a protected port
    trunk        Set trunking characteristics of the interface
    voice        Voice appliance attributes
Switch(config-if)#switchport trunk ?
    allowed      Set allowed VLAN characteristics when interface is in trunking
mode
    native       Set trunking native characteristics when interface is in trunking
mode
Switch(config-if)#switchport trunk
```

Copy | Paste

Fig 4.2 - Cisco 2950-24: identical result - no 'encapsulation' option under 'switchport trunk'.

```
2 Gigabit Ethernet/IEEE 802.3 interface(s)

63488K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address: 0010.117C.1772
Motherboard assembly number: 73-5781-09
Power supply part number: 34-0965-01
Motherboard serial number: FOC0610048Z
Power supply serial number: DAB0609127D
Model revision number: C0
Motherboard revision number: A0
Model number: WS-C2950T-24
System serial number: FHK0610Z0WC

Cisco Internetwork Operating System Software
IOS (tm) C2950 Software (C2950-I6Q4L2-M), Version 12.1(22)EA4, RELEASE
SOFTWARE(fc1)
Copyright (c) 1986-2005 by cisco Systems, Inc.
Compiled Wed 18-May-05 22:31 by jharirba

Press RETURN to get started!

Switch>ena
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#switchport trunk ?
    allowed  Set allowed VLAN characteristics when interface is in trunking
mode
    native   Set trunking native characteristics when interface is in trunking
mode
Switch(config-if)#switchport trunk
```

Copy

Paste

Fig 4.3 - Cisco 2950T-24: same limitation confirmed.

On the Cisco 3560-24, the 'encapsulation' keyword does exist, but it only accepts 'dot1q':

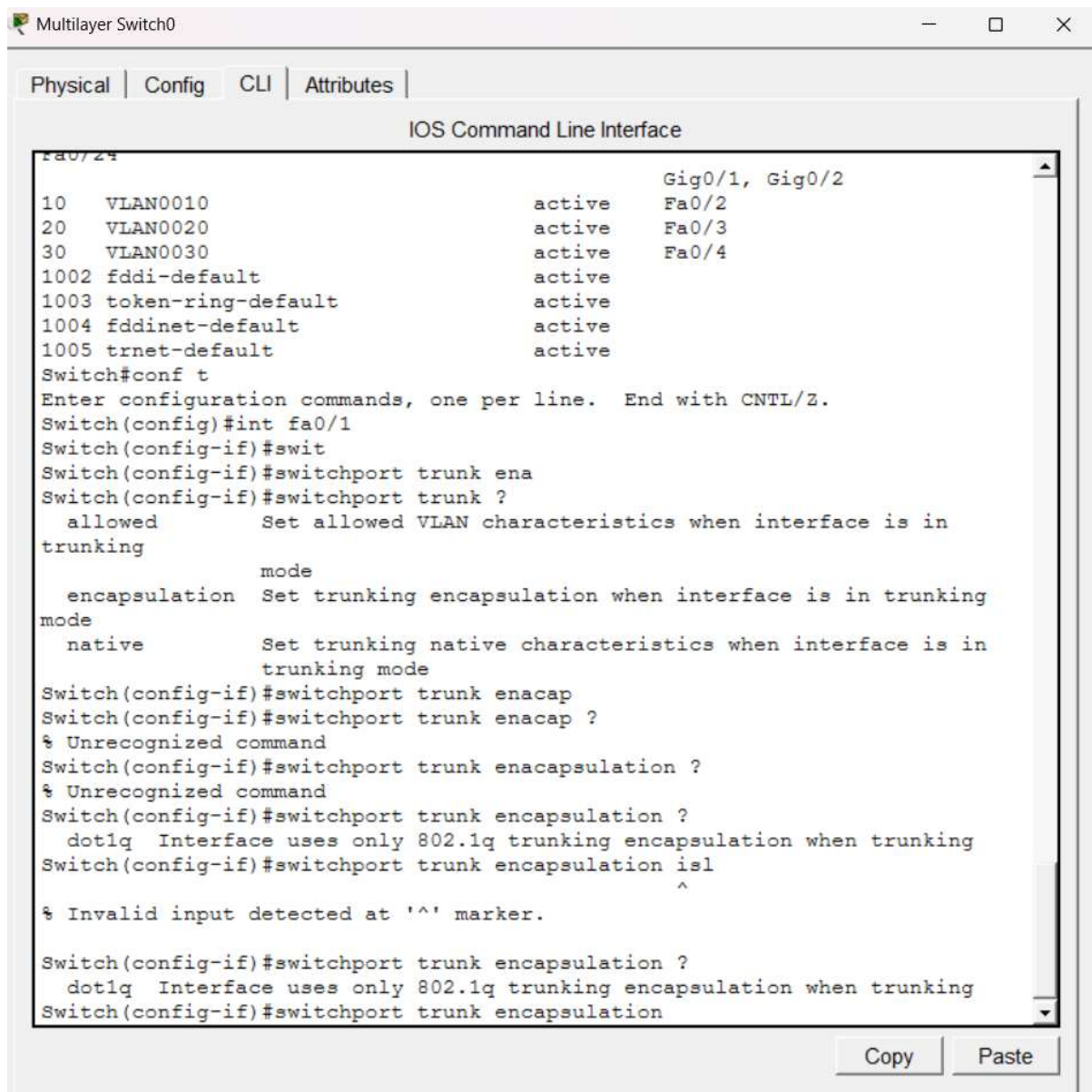


Fig 4.4 - Cisco 3560-24: 'switchport trunk encapsulation ?' shows only 'dot1q' is supported. Attempting 'switchport trunk encapsulation isl' returns '% Invalid input detected'.

Observation: ISL (Inter-Switch Link) could not be configured on any of the four switch models tested. On the 2960, 2950-24 and 2950T-24, the 'encapsulation' sub-command is entirely absent from the CLI. On the 3560-24, the sub-command exists but explicitly rejects 'isl' as invalid input, accepting only 'dot1q'.

Task 5: Simulate a scenario where a new port on Switch2 (FastEthernet0/5) is accidentally set as a trunk port instead of an access port for VLAN 20. Observe and explain the impact on connectivity for a PC connected to that port.

A new port, Fa0/5 on Switch1, was configured as a trunk port to simulate the accidental misconfiguration described in the task:

```
Switch(config)# interface fastEthernet 0/5
```

```
Switch(config-if)# switchport mode trunk
```


Initial attempt failed, since the 3560's default trunk encapsulation is 'Auto':

```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastEthernet 0/5
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto"
can not be configured to "trunk" mode.
Switch(config-if)#
```

Fig 5.1 - 'Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.'

This was resolved by explicitly setting the encapsulation before enabling trunk mode:

```
Switch(config-if)# switchport trunk encapsulation dot1q
```

```
Switch(config-if)# switchport mode trunk
```

A new PC (PC6) was connected to Fa0/5 with IP address 192.168.1.17 /24, as if it were intended to be an access port on VLAN 20:

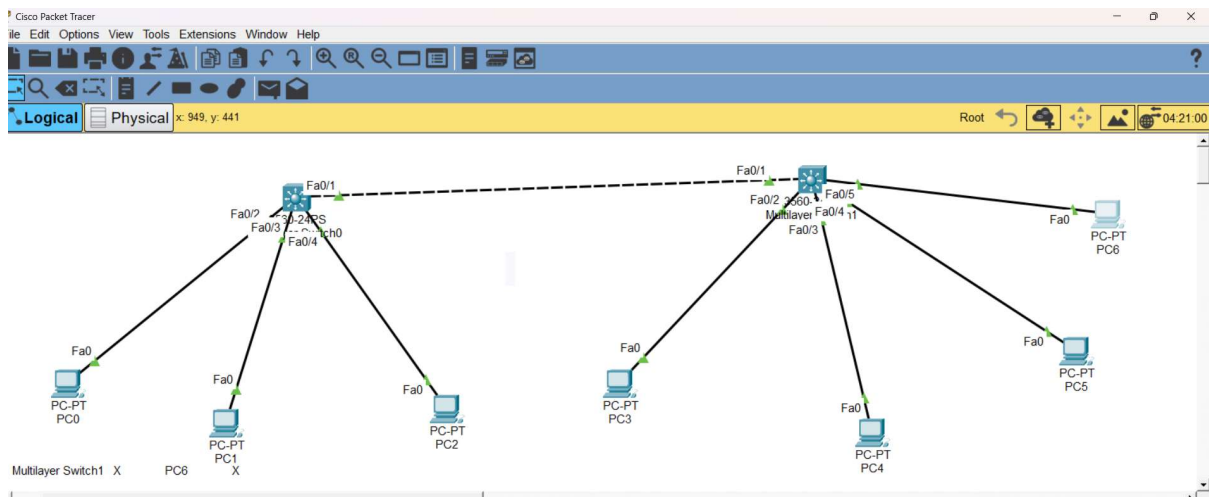


Fig 5.2 - PC6 connected to the misconfigured Fa0/5 trunk port on Switch1.

PC6 was then used to ping PC1 (192.168.1.12) and PC4 (192.168.1.15), both genuine VLAN 20 access-port devices:

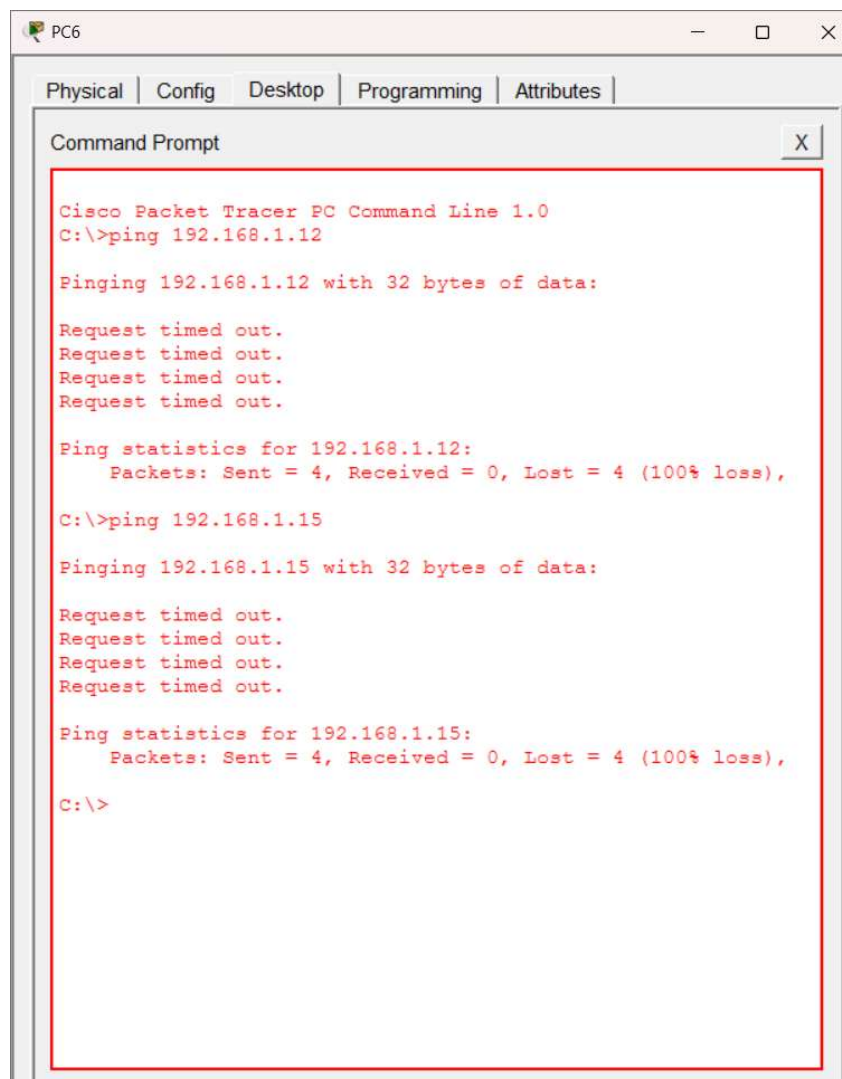


Fig 5.3 - Both pings from PC6 failed with 100% packet loss.

Observation: Connectivity failed completely (100% packet loss) for PC6, even though the physical link to Fa0/5 was up and no cabling issue existed.

Explanation: A trunk port does not behave like an access port. An access port assigns all connected traffic to a single, specific VLAN, whereas a trunk port expects incoming frames to already be 802.1Q-tagged with a VLAN ID. A standard PC network interface, such as PC6's, only ever sends plain untagged Ethernet frames - it has no concept of VLAN tagging. When these untagged frames arrive on a trunk port, the switch treats them as belonging to the trunk's native VLAN (VLAN 1 by default), not VLAN 20 as intended. Since no other VLAN 20 devices exist on native VLAN 1, and PC6 was never actually placed into VLAN 20's broadcast domain, its traffic never reaches PC1 or PC4. This demonstrates a realistic and common misconfiguration: mistakenly leaving a port in trunk mode instead of properly setting it to access mode with 'switchport access vlan 20' results in a device that appears physically connected, yet is functionally isolated from its intended network.